



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Student Course Registration System	
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Abstract:

Educational institutions manage course registration through systems that are often tightly coupled and monolithic in design, embedding prerequisite checks, seat allocation, and student notifications within a single application layer. This architecture makes the system difficult to extend, integrate with other university systems, or scale independently during peak enrollment periods. Service-Oriented Architecture (SOA) offers a solution by decomposing such systems into discrete, well-defined, and reusable services that communicate through standardized contracts, enabling loose coupling and interoperability across heterogeneous platforms.

To address this need, the proposed system restructures course registration into three independent SOAP-based web services, each defined by a WSDL contract, registered with a UDDI-style service registry, and coordinated through an Enterprise Service Bus (ESB) using orchestration logic.

The first service, the Prerequisite and Conflict Validation Service (PCVS), exposes prerequisite checking and timetable conflict detection as an XML/SOAP-based operation, invoked by the ESB before any enrollment request proceeds. Decoupling this logic from the core registration application allows validation rules to be updated independently without redeploying the entire system.

The second service, the Seat Allocation Service (SAS), manages real-time seat availability per course section, exposing operations for checking, reserving, and releasing seats. Structuring this as its own service prevents allocation logic from being duplicated across other parts of the system and allows it to be reused by other consumers (e.g., a departmental override tool) through the same WSDL contract. The third service, the Notification Service (NS), is invoked by the ESB whenever an enrollment is confirmed, waitlisted, or a seat becomes available, sending alerts to students through a standardized SOAP interface, independent of how the notification is ultimately delivered (email, SMS, or portal alert).

An orchestration layer, implemented using BPEL-style process logic within the ESB, sequences calls to PCVS, SAS, and NS to complete an end-to-end registration transaction. This design demonstrates the core SOA principles of loose coupling, service reusability, standardized contracts, and platform interoperability, and provides a foundation that can be extended to cross-departmental or cross-institutional registration scenarios where multiple universities expose their own course services under a shared registry.

Project Architecture/Project Overview:

